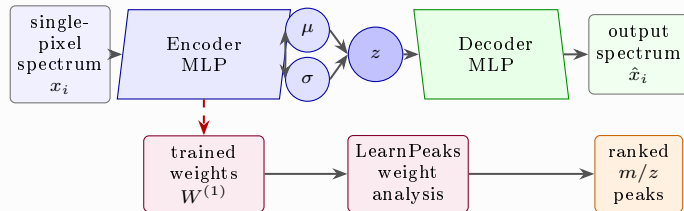


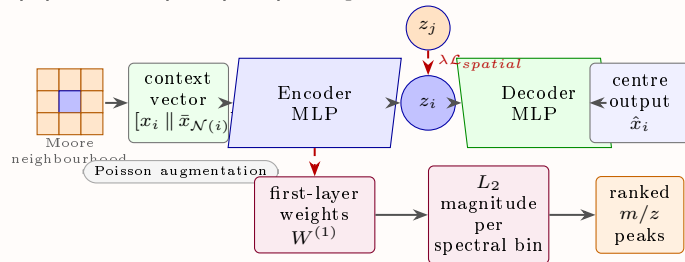
(A) Legacy msiPL

spatially blind spectral VAE



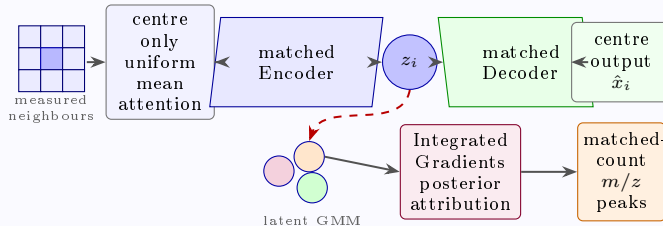
(B) Proposed Spatial-msiPL

proposal: context plus explicit spatial regularisation



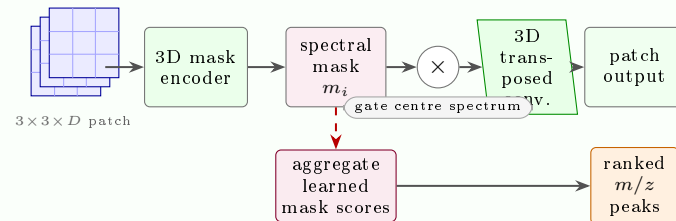
(C) Evaluated Spatial-msiPL

matched VAE ablation with nonlinear attribution



(D) S3PL

spatial-spectral self-supervision with 3D convolutions



Shared evaluation: selected m/z bins $\rightarrow$ ion images $\rightarrow$ PCC against expert masks $\rightarrow$ mSCF1